

# VARIANCE-COVARIANCE ESTIMATION WITH SPATIAL AND SERIAL CORRELATION: IV IN AN UN-BALANCED PANEL

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This version: June 20, 2018

## 1 What I do

I provide the adjusted formulas for, and analyze the behavior of, five estimators for the SE of a pooled OLS-IV estimation in the case of unbalanced panels. The five estimators I consider are: (i) heteroskedastic-robust ("White"); (ii) two single-cluster estimators (by firm and by time, separately); (iii) a 2-way clustering estimator; and (iv) the "average of HACs" estimator proposed by Driscoll and Kraay (1998), and extended to the unbalanced-panel case by Hoechle (2016).

## 2 Insights and summary of findings

- **The "ham" of the sandwich, and experimental design:** the core of all the SE estimators (i.e. the "ham" in the sandwich formula) contains variances and covariances that depend on the distribution of **both** the regressor(s) and error term. It is therefore crucial to understand what is the sampling /experimental design behind the data sample at hand. In our firm-level paper, we know that the regressor (and instrument) are both **common** to all firms within a single period, and probably uncorrelated over time. Unless the error term is i.i.d. (something quite improbable, and indeed, something that does not hold within a LP estimation), then this already hints to a need to cluster (at least) along the time dimension. Simulation results confirm this.
- **Unequal size of panel dimensions:** Whenever the two panel dimensions have very different sizes, with one dimension being very large and the other relatively small (in our case, time much smaller than the firm dimension), then moving from no-clustering (White robust) to clustering along the large dimension (firms in our case) will have small effects on the SE estimates, although it might

risk a significant bias if it turns out that there is within-time correlation (i.e. correlation across firms at a given point in time) in regressors and/or errors. On the other hand, not clustering when there is indeed correlation along the smaller dimension, will imply a significant bias in the SE estimates. In other words, moving from no-clustering to clustering along the small dimension (time) will have a significant effect if there is indeed within-time correlation. In this sense, *everything else equal*, it might be advisable to cluster along the smaller dimension. One has to bear in mind, however, that if the number of clusters is "too small", then this might imply a poor finite-sample estimate.

- **Behavior of the 2-way cluster estimator:** In the benchmark simulations with a regressor **common** to all firms, with large  $N$  and small  $T$  and both serial and spatial correlation (similar to the situation we are probably dealing with in our paper), the 2-way clustering estimator has very similar median and coverage rates as the time-clustering estimator, mainly due to the reasons explained above. See panels C and D of table 1, as well as panels B and C of table 2). It does present, however, a larger variance; this is also the case for the Driscoll-Kraay estimator, and a feature present in more general distributions of the regressor (see the figures with the distribution of the SE estimates). As argued by ?, noisy SE estimates might lead to wrong inference, due to a Jensen's inequality type of argument. On the other hand, and in line with the recent literature, my results here show that 2-way clustering is preferred relative to one-way clustering in a setting with panel dimensions that are relatively similar (and not too large) in size; see panel D of table 2).
- **Behavior of the Driscoll-Kraay estimator:** the behavior of this estimator is very similar to the 2-way clustering estimator. Both rely on  $T$ -asymptotics. The main difference arises when there is non-contemporaneous spatial correlation, i.e. different firms correlated across periods. In such case, the simple 2-way clustering estimator I use here is inconsistent, while DK is still consistent. This can be seen comparing panels B and C of table 2).
- **True vs. aggregated data frequency:** when comparing estimations using the "true" monthly frequency from the DGP (columns 1 and 3 in tables 1-2) versus using aggregated annual data (columns 2 and 4 in tables 1-2), some features emerge. First, and as expected, the true SE of  $\beta^{IV}$ , as well as the variance of the SE estimates, increases when moving from true to aggregated data. Second, and more interesting, the behavior of the White and firm-cluster estimators *improves* with aggregated data, while the behavior of the time-clustering, 2-way clustering and Driscoll-Kraay estimators worsens.

### 3 Set-up and derivations

We have a smple  $(y_{i,1}, \dots, y_{i,T_i}; x_{i,1}, \dots, x_{i,T_i}, \alpha_i)$ ,  $i = 1, \dots, N$  from the following population model

$$y_{i,t} = x'_{i,t}\beta + \alpha_i + \epsilon_{i,t} \quad (1)$$

where

- the total number of periods  $T_i$  can vary accross individual firms  $i$ ; i.e. we have a potentially **un-balanced** panel,
- $x_{i,t}$  is a  $k \times 1$  vector of potentially endogenous variables
- there is a  $r \times 1$  vector of instruments  $z_{i,t}$ , with  $r \geq k$ , such that

$$E(\epsilon_i \mid \mathbf{z}_i) = 0 \quad (2)$$

where  $\epsilon_i = (\epsilon_{i,1}, \dots, \epsilon_{i,T_i})'$ ,  $\mathbf{z}_i = (z_{i,1}, \dots, z_{i,T_i})'$ .

Assumption (2) is the *strict exogeneity* assumption often made (explicitly or implicitly) in these situations, although for variable(s) in  $x$ . It implies that  $z$  at any period is uncorrelated with past, present and future values of  $\epsilon$ . In particular, assumption (2) implies the following  $r \times 1$  orthogonality conditions that can be used in a GMM-type estimation:

$$E(z_{i,t}(\tilde{y}_{i,t} - \tilde{x}'_{i,t}\beta)) = 0 \quad (3)$$

where  $\tilde{y}_{i,t} = y_{i,t} - \bar{y}_i$  and  $\tilde{x}_{i,t} = x_{i,t} - \bar{x}_i$  are the *within-group* transformations, and the expectations are taken with respect to the distribution of the cross section.

We are interested in the properties of an estimator for  $\beta$  when the following standard assumption **does not** hold:

$$Var(\epsilon_i \mid \mathbf{x}_i, \alpha_i) = \sigma^2 I_T \quad (4)$$

For the rest of the exposition, lets assume that  $r = k$ ; i.e. there is one instrument for each endogenous variable. Stacking all observations such that  $Z = (\mathbf{z}'_1, \dots, \mathbf{z}'_N)'$  is a  $(T_1 + \dots + T_N) \times k$  matrix, the orthogonality condition (3) allows us to exppress the IV-estimate of  $\beta$  as

$$\hat{\beta}^{IV} = (Z' \tilde{X})^{-1} Z' \tilde{Y} \quad (5)$$

Under standard regularity conditions, and given that

$$\hat{\beta}^{IV} - \beta = \left( \sum_{i=1}^N \sum_{t=1}^{T_i} z_{i,t} \tilde{x}'_{i,t} \right)^{-1} \sum_{i=1}^N \sum_{t=1}^{T_i} z_{i,t} \tilde{\epsilon}_{i,t}$$

with  $E(\mathbf{z}_i' \tilde{\boldsymbol{\epsilon}}_i) = 0$ , a CLT can be applied assuming  $T_i \rightarrow \infty$  when  $T \rightarrow \infty$ ,  $\forall i$

$$\left( \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^{T_i} z_{i,t} \tilde{\mathbf{x}}_{i,t}' \right) \sqrt{d_{NT}} \left( \hat{\beta}^{IV} - \beta \right) \xrightarrow{d} \mathcal{N}[0, W_0]$$

where  $d_{NT}$  is the speed of convergence, and

$$\begin{aligned} W_0 &= \lim_{N, T \rightarrow \infty} \frac{1}{NT} \text{Var} \left( \sum_{i=1}^N \sum_{t=1}^{T_i} z_{i,t} \tilde{\mathbf{x}}_{i,t} \right) \\ &= \lim_{N, T \rightarrow \infty} \frac{1}{NT} \left[ \sum_{i=1}^N \sum_{t=1}^{T_i} \text{Var} (z_{i,t} \tilde{\mathbf{x}}_{i,t}) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \sum_{t=1}^{T_i} \sum_{s=1}^{T_j} \text{Cov} (z_{i,t} \tilde{\mathbf{x}}_{i,t}, z_{j,s} \tilde{\mathbf{x}}_{j,s}) \right] \end{aligned} \quad (6)$$

From (6) above, it is clear that three features of the DGP will be crucial to understand the behavior of different estimators: (i) the distribution of  $z_{i,t}$ , (ii) the distribution of  $\tilde{\mathbf{x}}_{i,t}$ , and (iii) the relative sizes of  $N$  and  $T$ . Different assumptions about (i) and (ii) will lead to different expressions and “reductions” in (6).

Applying standard Slutsky and Cramer theorems, and as long as  $\text{plim} \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i \tilde{\mathbf{x}}_i' = E(\mathbf{z}_i \tilde{\mathbf{x}}_i')$  with full rank  $= k$

$$\sqrt{d_{NT}} \left( \hat{\beta}^{IV} - \beta \right) \xrightarrow{d} \mathcal{N} \left[ 0, (E(\mathbf{z}_i \tilde{\mathbf{x}}_i'))^{-1} W_0 (E(\mathbf{z}_i \tilde{\mathbf{x}}_i'))^{-1} \right] \quad (7)$$

What we are after here is a consistent estimate of  $W_0$ , the “ham” in the sandwich formula (7).

### 3.1 No Spatial/Serial correlation: White (1980)’s estimator

If we assume that observations are spatially uncorrelated, but allow for general heteroskedasticity and autocorrelation, the off - (block) diagonal terms (i.e. the covariances) in (6) disappear, and we can then use the standard HAC-robust estimator

$$\widehat{W}_{White} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i \tilde{\boldsymbol{\epsilon}}_i \tilde{\boldsymbol{\epsilon}}_i' \mathbf{z}_i' \quad (8)$$

A consistent estimate for the asymptotic variance of  $\beta^{IV}$  is then given by

$$\widehat{\text{Var}} \left( \hat{\beta}^{IV} \right) = \left( Z' \tilde{X} \right)^{-1} \left( \sum_{i=1}^N \mathbf{z}_i \tilde{\boldsymbol{\epsilon}}_i \tilde{\boldsymbol{\epsilon}}_i' \mathbf{z}_i' \right) \left( \tilde{X}' Z \right)^{-1} \quad (9)$$

### 3.2 Spatial - Serial correlation robust

If instead we allow for general forms of **both** spatial and serial correlation, then the off-diagonal elements (i.e. the co-variances) in (6) are potentially different from 0. Several estimators have been proposed for such case, although they rely on different assumptions about the cross-unit-time correlations.

#### 3.2.1 Two-Way Cluster Estimator

To make the argument clear, let's make the following assumption about non-contemporaneous spatial and time dependence

$$\begin{aligned} E(z_{i,t}\tilde{\epsilon}_{i,t}\tilde{\epsilon}'_{j,t}z'_{j,t}) &\neq 0 \\ E(z_{i,t}\tilde{\epsilon}_{i,t}\tilde{\epsilon}'_{i,s}z'_{i,s}) &\neq 0 \\ E(z_{i,t}\tilde{\epsilon}_{i,t}\tilde{\epsilon}'_{j,s}z'_{j,s}) &= 0, \forall i \neq j, \forall t \neq s \end{aligned}$$

Under the above assumptions, adding and subtracting  $\sum_{i=1}^N \sum_{t=1}^{T_i} Var(z_{i,t}\tilde{\epsilon}_{i,t})$ , expression (6) can be written as

$$W_0 = \lim_{N,T \rightarrow \infty} \left[ \frac{1}{NT} \sum_{i=1}^N E(W_i) + \frac{1}{NT} \sum_{t=1}^T E(W_t) - \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T E(z_{i,t}\tilde{\epsilon}_{i,t}\tilde{\epsilon}'_{i,t}z'_{i,t}) \right] \quad (10)$$

where

$$\frac{1}{NT} \sum_{i=1}^N E(W_i) \equiv W_{Cfirm} = \frac{1}{N} \sum_{i=1}^N \left( \frac{1}{T} \sum_{t=1}^{T_i} \sum_{s=1}^{T_i} E(z_{i,t}\tilde{\epsilon}_{i,t}\tilde{\epsilon}'_{i,s}z'_{i,s}) \right) \quad (11)$$

$$\frac{1}{NT} \sum_{t=1}^T E(W_t) \equiv W_{Ctime} = \frac{1}{T} \sum_{t=1}^T \left( \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N E(z_{i,t}\tilde{\epsilon}_{i,t}\tilde{\epsilon}'_{j,t}z'_{j,t}) \right) \quad (12)$$

Then, following Cameron et al. (2011) and Thompson (2011), I can define the 2-way clustering estimator as

$$\widehat{Var}(\hat{\beta}^{IV}) = (Z' \tilde{X})^{-1} \widehat{W}_{2CCE} (\tilde{X}' Z)^{-1} \quad (13)$$

where  $W_{2CCE}$  is the sample analog of expression (10)

$$\widehat{W}_{2CCE} = \widehat{W}_{Cfirm} + \widehat{W}_{Ctime} - \widehat{W}_{White} \quad (14)$$

where  $\widehat{W}_{Cfirm} = \frac{1}{NT} \sum_{i=1}^N \widehat{W}_i$ , and  $\widehat{W}_{Ctime} = \frac{1}{NT} \sum_{t=1}^T \widehat{W}_t$ .

In other words, (14) is the sum of firm and time cluster estimators, minus the heteroskedasticity-robust estimator defined in (8). Note that (14) is a consistent estimator of  $W_0$  when there is both correlation across firms, and within firm across time, but there is no non-contemporaneous spatial correlation.

A couple of additional points, which will be useful to understand the simulation results below:

- As pointed above, the relevant correlation is between different values of  $z_{i,t}\tilde{\epsilon}_{i,t}$ . This means that, for example, even if  $\tilde{\epsilon}_{i,t}$  exhibits correlation within a firm, if the instrument  $z_{i,t}$  is **not** correlated across time, then firm-clustering component  $\widehat{W}_{Cfirm}$  will not affect the SE.
- As argued in Thompson (2011), if the  $N$  and  $T$  dimensions are very unbalanced, everything else equal, it is "safer" (i.e. chances of a biased estimate are lower) to cluster along the dimension with *fewer* observations. The problem with this, however, is that if the number of clusters is relatively small, then even when the estimator is consistent, it might exhibit large dispersion in small samples. By means of a Jensen's inequality argument, Thompson (2011) shows that an increase of the dispersion of the SE estimator might lead to "over-reject" the null of  $\beta^{IV} = 0$ .
- If one suspects that the true DGP exhibits non-contemporaneous spatial correlation, then (14) is not a consistent estimator of (6).

### 3.2.2 Driscoll and Kraay (1998) Estimator

We then rely on the results in Driscoll & Kraay (1998), modified along the lines of Hoechle (2016) to allow for unbalanced panels. Given the derivations above, it is straightforward to apply these estimators to our set-up with IV, and get the following expression for a consistent estimator

$$\widehat{Var}(\hat{\beta}^{IV}) = (Z' \tilde{X})^{-1} \hat{S}_T (\tilde{X}' Z)^{-1} \quad (15)$$

where  $\hat{S}_T$  is given by the adapted Newey and West (1987) estimator

$$\hat{S}_T = \hat{\Omega}_0 + \sum_{j=1}^{m(T)} \phi(j, m(T)) (\hat{\Omega}_j + \hat{\Omega}'_j) \quad (16)$$

In the expression above,  $m(T)$  is the lag-length truncation for the autocorrelation of residuals; the Bartlet weights being

$$\phi(j, m(T)) = 1 - \frac{j}{m(T) + 1}$$

and the  $k \times k$  matrix  $\hat{\Omega}_j$  given by

$$\hat{\Omega}_j = \sum_{t=j+1}^T z_t \tilde{\epsilon}_t \tilde{\epsilon}'_{t-j} z'_{t-j} \quad (17)$$

with

$$z_t \tilde{\epsilon}_t = \sum_{i=1}^{N(t)} z_{i,t} \tilde{\epsilon}_{i,t}$$

In the Monte Carlo experiment below, I will compare the estimators (9) and (15).

## 4 Monte Carlo experiment

For the simulations below, I modify the original experiments in Driscoll and Kraay (1998) in order to have a set-up as close as possible to the one we face in Cloyne et al. (2018), both in terms of potential "true" DGP and sampling. In particular, (i) I abstract from unobserved heterogeneity (i.e.  $\alpha_i = 0 \ \forall i$ ); (ii) I only allow for an aggregate RHS variable  $x$  and instrument  $z$ , which present no auto-correlation; and (iii) I use both a "true" and an aggregated data frequency.

### 4.1 True DGP

Let the benchmark population model, which somehow mimics properties of what we estimate in Cloyne et al. (2018), be

$$y_{i,t} = x_t \beta + \epsilon_{i,t} \quad (18)$$

where  $x_t$  is a single variable (i.e.  $k = 1$ ),  $t$  represents **months**. The (potentially) endogenous variable  $x_t$  and the *instrument*  $z_t$  are defined such that  $\text{corr}(x, z) = \rho_z$

$$x_t = \rho_z z_t + \sqrt{1 - \rho_z^2} w_t \quad (19)$$

where  $w_t \sim (0, \sigma_w^2)$ . In other words,

$$\begin{aligned} z_{i,t} &= z_t \ \forall i \\ z_t &\perp z_s \ \forall t \neq s \end{aligned}$$

The error term has the following structure

$$\epsilon_{i,t} = \lambda_i f_t + \eta_{i,t} + u_{i,t} \quad (20)$$

where

$$\begin{aligned} f_t &= \rho_f f_{t-1} + v_t \\ \eta_{i,t} &= \rho_\eta \eta_{i,t-1} + e_{i,t} \end{aligned}$$

and the factor loadings  $\lambda_i \stackrel{iid}{\sim} U[0, b]$ . I normalize the variance of  $\epsilon_{i,t}$  so that these are homoskedastic. This can be achieved by assuming

$$\begin{aligned} E(u_{i,t}^2) &= 1 - \lambda_i^2 \\ E(v_t^2) &= 1 - \rho_f^2 \\ E(e_{i,t}^2) &= 1 - \rho_\eta^2 \end{aligned}$$

The above then implies

I then pick values for the  $\rho$ 's and  $b$  in order to control the spatial and serial correlation.

## 4.2 Sampling

I draw  $S$  samples, each sample consisting of  $(y_{i,1}, \dots, y_{i,T_i}; x_1, \dots, x_T; z_1, \dots, z_T), i = 1, \dots, N$ , with  $T_i \leq T$ . In the benchmark simulations, I set  $S = 1000, N = 2000, T = 20 \times 12$  months, and I select the time frame for each firm  $i$  in the following way:

$$T_i \stackrel{iid}{\sim} U \left[ \frac{1}{2}T, T \right] \quad (21)$$

which implies that all firms are observed for at least  $\text{ceil}(\frac{1}{2}T)$  periods, after which their final observation is uniformly *iid* distributed.

I will consider estimations from two time frequencies:

1. **“True” frequency:** This is the benchmark, monthly frequency, with  $T = 20 \times 12$ .
2. **“Aggregated” frequency:** in this case, I consider **annual** frequency, such that  $\tilde{T} = 20$ . In this case, I assume that each firm randomly selects in which month to report, and this is denoted by  $d_i$ . I then aggregate the variables  $x, y, z$  into  $\{x_{d_i, \tilde{t}}, y_{d_i, \tilde{t}}, z_{d_i, \tilde{t}}\}$  such that, for  $\tilde{t} = 2, \dots, \tilde{T}$ , each observation corresponds to the sum of the previous 12 months considering the reporting month  $d_i$  of each firm

$$\begin{aligned} x_{d_i, \tilde{t}} &= \sum_{s=1}^{12} x_{12(\tilde{t}-1)+d_i, s} \\ z_{d_i, \tilde{t}} &= \sum_{s=1}^{12} z_{12(\tilde{t}-1)+d_i, s} \\ y_{d_i, \tilde{t}} &= \sum_{s=1}^{12} y_{i, 12(\tilde{t}-1)+d_i, s} \end{aligned}$$

I then run a panel regression at the annual frequency.

The sequence in the simulation is the following: (i) draw the  $\lambda$ 's for each firm, together with the idiosyncratic firm horizons  $T_i$ ; these will stay *fixed* throughout the different random draws (ii) generate  $S$  draws for  $\{x, y, z\}$  and estimate the regression model with the two alternative frequencies (iii) store the estimated  $\beta$ 's and the estimated SE's. I will focus on five alternative SE estimators, which will be consistent or not for the true SE depending on the DGP: (i) heteroskedastic-robust (“White”); (ii) firm-cluster; (iii) time-cluster; (iv) 2-way cluster; and (v) Driscoll-Kraay.

## 4.3 Results

I present findings using, for each SE estimator, three statistical objects: (i) the ratio estimated-to-empirical SE, where the numerator is the median of the SE distribution



while the empirical is computed from the distribution for  $\beta^{IV}$  generated from  $S$  replications; (ii) the 95% CI coverage ratio; and (iii) the entire distribution of each SE estimator as generated from the  $S$  replications. Statistics (i) and (ii) are presented in tables 1-2, while the distributions in (iii) are presented in figures 1-??.

#### 4.3.1 Benchmark scenario

All benchmark simulations are run for the following set of sample parameters:  $T = 20$ ,  $N = 2000$ ,  $S = 1000$ , and I simulate data from the DGP (18) with a **common** explanatory variable  $x_t$ . I then consider four alternative assumptions about the error term, which are plausible scenarios in the empirical setting faced in Cloyne et al. (2018).

1. **No correlation:** this amounts to setting  $b = \rho_\eta = 0$
2. **Only serial correlation:** this amounts to setting  $b = 0, \rho_\eta > 0$ . In particular, and in order to make the argument more explicit, I set  $\rho_\eta = 0.85$ .
3. **Serial and contemporaneous-spatial correlation:** this amounts to setting  $b > 0, \rho_\eta > 0, \rho_f = 0$ .
4. **Serial and spatial correlation:** this amounts to setting  $b > 0, \rho_\eta > 0, \rho_f > 0$ . In this case, I follow ? and set  $\rho_f = 0.25$  in order to have positive but weak non-contemporaneous spatial correlation.

Table 1, and figures 1-4 present the results. In table 1, the rows of each panel represent the 5 alternative SE estimators; the four columns present, for the estimations using the true frequency (columns 1 and 3) and the aggregated frequency (columns 2 and 4), the ratio of the estimated (mean)-to-true SE and the 95% nominal coverage rate.

#### 4.3.2 Alternative scenarios

In order to better understand the behavior of the different estimators, I consider four alternative scenarios (i.e. assumptions about the DGP and/or available sample). In particular, I now run simulations where I allow the regressor in (18) to vary over firms:

$$y_{i,t} = x_{i,t}\beta + \epsilon_{i,t} \quad (22)$$

where the error term  $\epsilon_{i,t}$  follows the same distribution as above, while now I also allow the regressor  $x_{i,t}$  to have a spatial and serial component:

$$x_{i,t} = f_t^x + \eta_{i,t}^x + u_{i,t}^x \quad (23)$$

with

$$\begin{aligned} f_t^x &= \rho_f^x f_{t-1}^x + v_t^x \\ \eta_{i,t}^x &= \rho_\eta^x \eta_{i,t-1}^x + e_{i,t}^x \end{aligned}$$

I then consider four scenarios which are characterized by different assumptions on both the regressor (and instrument) and error correlation structure; the first three scenarios (panels A,B and C in table 2) use the same panel dimensions as above  $T = 20, N = 2000$  while the last scenario uses  $T = 20, N = 50$ :

1. **Only serial correlation, in regressor and error:** this amounts to setting  $b = 0, \rho_\eta > 0, \rho_\eta^x > 0$ . In particular, and in order to make the argument more explicit, I set  $\rho_\eta = \rho_\eta^x = 0.85$ .
2. **Serial and (weak) non-contemporaneous spatial correlation:** I set  $b = 1, \rho_\eta = \rho_\eta^x = 0.85, \rho_f = \rho_f^x = 0.25$ .
3. **Serial and (strong) non-contemporaneous spatial correlation:** I set  $b = 1, \rho_\eta = \rho_\eta^x = 0.85, \rho_f = \rho_f^x = 0.5$ .
4. **Serial and (weak) non-contemporaneous spatial correlation with small panel:** This is similar to scenario 2 above, wexcept that now I use a relatively small panel with  $T = 20, N = 50$ .

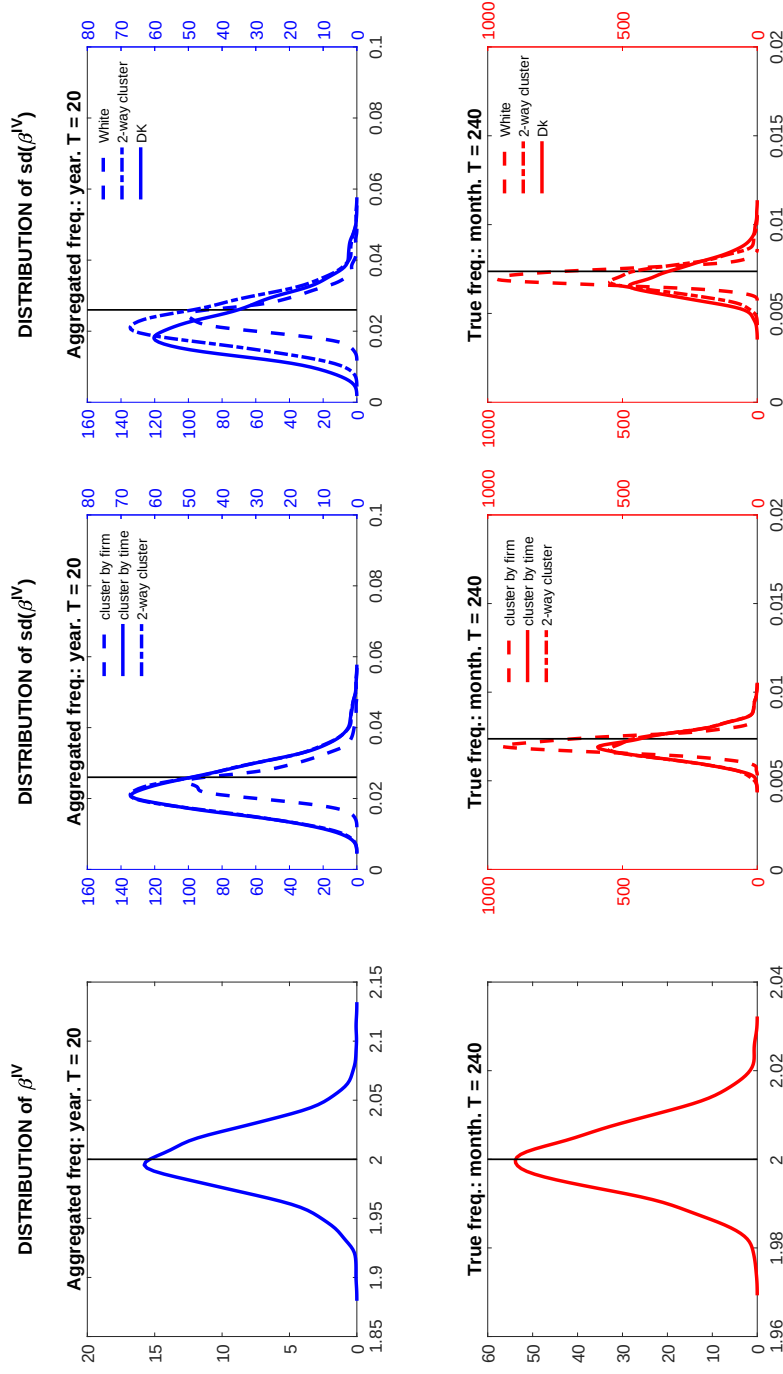
Table 2 and figures 5-8 present the results for these four alternative scenarios. The structure of the table and figures is the same as in the benchmark results above.

Table 1: PERFORMANCE OF DIFFERENT ESTIMATORS FOR  $SE(\hat{\beta})$   
BENCHMARK: **common regressor**,  $x_t$ , WITH DIFFERENT ASSUMPTIONS FOR  
ERROR TERM.  $N = 2000$  AND MAX  $\tilde{T} = 20$  YEARS

| ESTIMATOR                                                | ESTIMATED-TO-EMPIRICAL SE |            | 95% CI COVERAGE RATE |            |
|----------------------------------------------------------|---------------------------|------------|----------------------|------------|
|                                                          | True frequency            | Aggregated | True frequency       | Aggregated |
| PANEL A: NO CORRELATION                                  |                           |            |                      |            |
| H-robust                                                 | 0.99                      | 1.00       | 0.95                 | 0.95       |
| Firm cluster                                             | 0.99                      | 1.00       | 0.95                 | 0.95       |
| Time cluster                                             | 0.99                      | 1.00       | 0.95                 | 0.95       |
| 2-way cluster                                            | 0.99                      | 1.00       | 0.95                 | 0.95       |
| Driscoll-Kraay                                           | 0.98                      | 0.90       | 0.95                 | 0.89       |
| PANEL B: SERIAL CORRELATION ONLY                         |                           |            |                      |            |
| H-robust                                                 | 0.96                      | 0.97       | 0.94                 | 0.94       |
| Firm cluster                                             | 0.96                      | 0.98       | 0.94                 | 0.95       |
| Time cluster                                             | 0.94                      | 0.88       | 0.93                 | 0.91       |
| 2-way cluster                                            | 0.94                      | 0.90       | 0.93                 | 0.90       |
| Driscoll-Kraay                                           | 0.93                      | 0.83       | 0.93                 | 0.89       |
| PANEL C: SERIAL/CONTEMPORANEOUS-SPATIAL CORR.            |                           |            |                      |            |
| H-robust                                                 | 0.09                      | 0.22       | 0.15                 | 0.36       |
| Firm cluster                                             | 0.09                      | 0.22       | 0.14                 | 0.35       |
| Time cluster                                             | 0.97                      | 0.74       | 0.94                 | 0.84       |
| 2-way cluster                                            | 0.98                      | 0.74       | 0.94                 | 0.83       |
| Driscoll-Kraay                                           | 0.96                      | 0.75       | 0.94                 | 0.85       |
| PANEL D: SERIAL/(WEAK) NON-CONTEMPORANEOUS SPATIAL CORR. |                           |            |                      |            |
| H-robust                                                 | 0.09                      | 0.18       | 0.14                 | 0.29       |
| Firm cluster                                             | 0.09                      | 0.18       | 0.13                 | 0.28       |
| Time cluster                                             | 0.98                      | 0.74       | 0.95                 | 0.95       |
| 2-way cluster                                            | 0.98                      | 0.74       | 0.95                 | 0.83       |
| Driscoll-Kraay                                           | 0.96                      | 0.75       | 0.94                 | 0.84       |

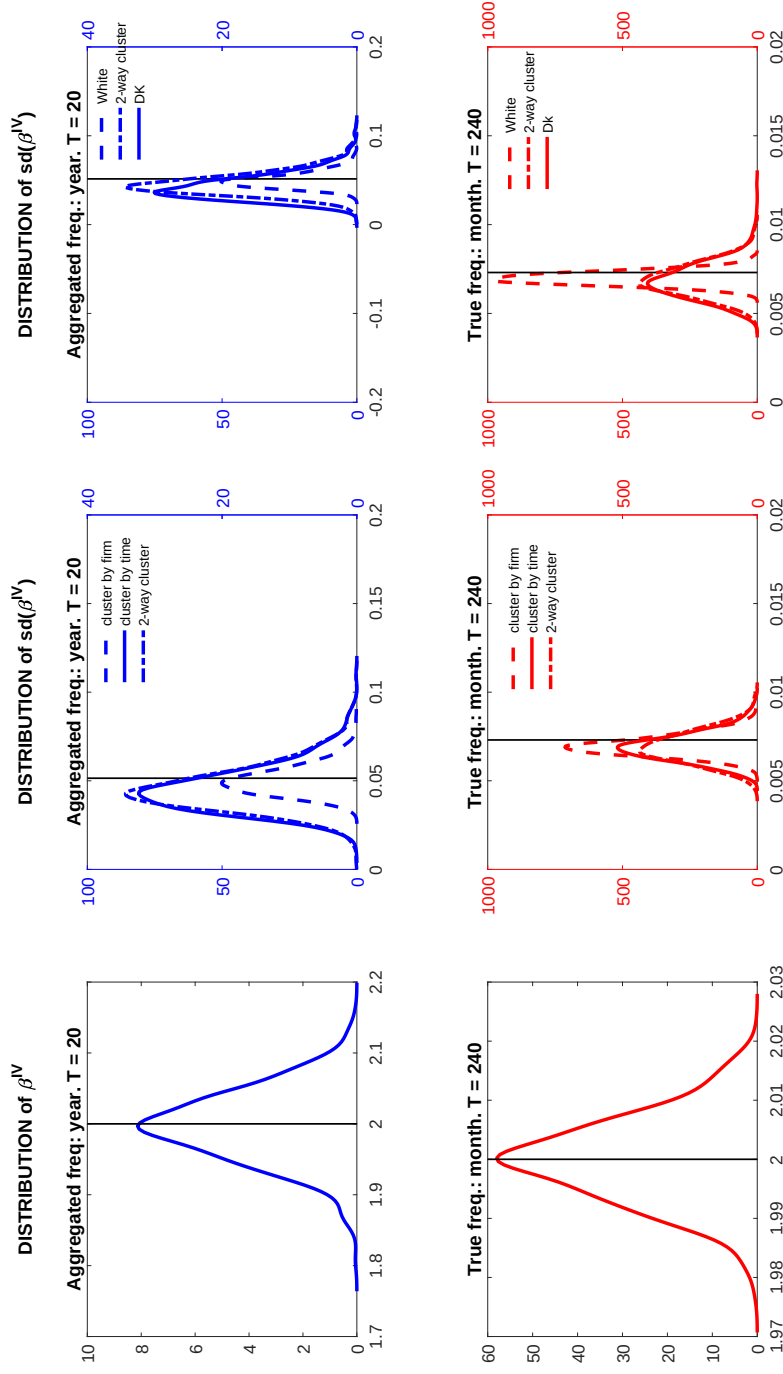
Notes: The estimated-to-empirical is computed as the ratio between the mean estimated SE using the estimators proposed and the mean SE computed from the simulated distribution of  $\beta^{IV}$ . The 95% coverage rate presents the percentage of simulation draws for which the estimated 95% confidence interval contained the true  $\beta$ . Each panel presents results for a different set of assumptions regarding the distribution of the error term in equation (18) of the main text.

Figure 1: ESTIMATED  $\beta$  AND SE. DGP WITH: COMMON REGRESSOR AND no Correlation in ERRORS.  $N = 2000, S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



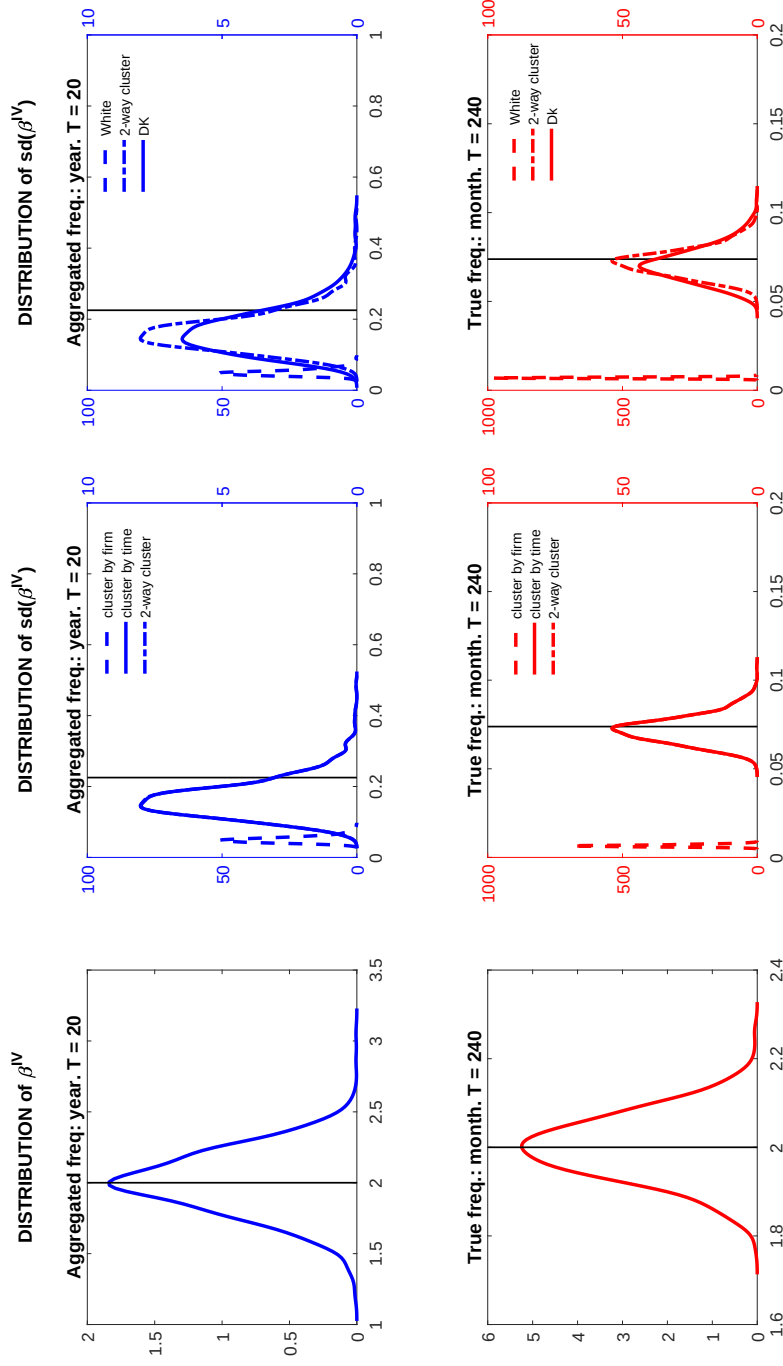
Note: this maps into panel A of table 1. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from the DGP (18). The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation are:  $\rho_\eta = b = 0$ .

Figure 2: ESTIMATED  $\beta$  AND SE WITH: COMMON REGRESSOR AND SERIAL CORRELATION IN ERRORS.  $N = 2000$ ,  $S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



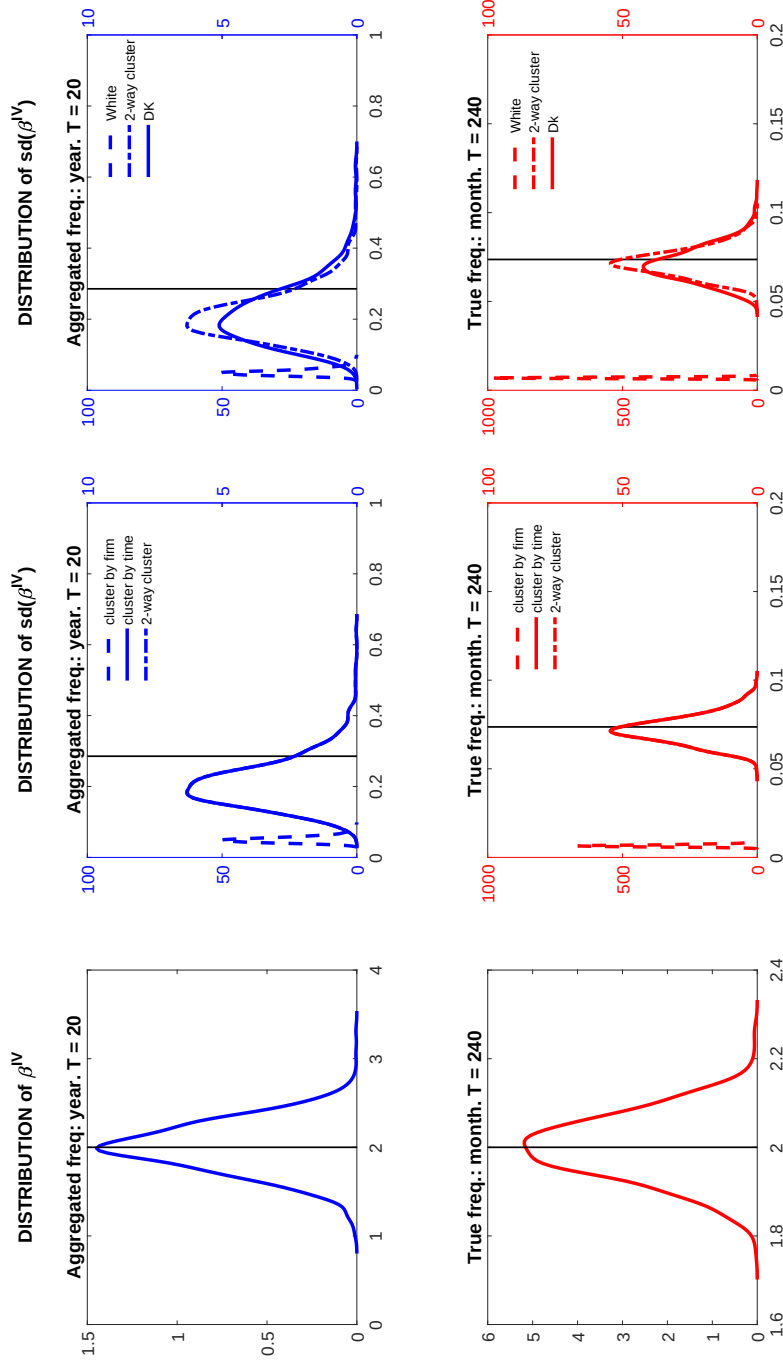
Note: this maps into panel B of table 1. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from the DGP (18). The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation are:  $\rho_\eta = 0.85$ ,  $b = 0$ .

Figure 3: ESTIMATED  $\beta$  AND SE WITH: COMMON REGRESSOR AND SERIAL/ SPATIAL CONTEMPORANEOUS CORRELATION IN ERRORS.  $N = 2000, S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



Note: this maps into panel C of table 1. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from the DGP (18). The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation are:  $\rho_\eta = 0.85, \rho_f = 0, b = 1$ .

Figure 4: ESTIMATED  $\beta$  AND SE WITH: COMMON REGRESSOR AND SERIAL/ SPATIAL CORRELATION IN ERRORS.  
 $N = 2000, S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



Note: this maps into panel D of table 1. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from the DGP (18). The top row shows results from aggregating the monthly frequency on the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation are:  $\rho_\eta = 0.85, \rho_f = 0.25, b = 1$ .

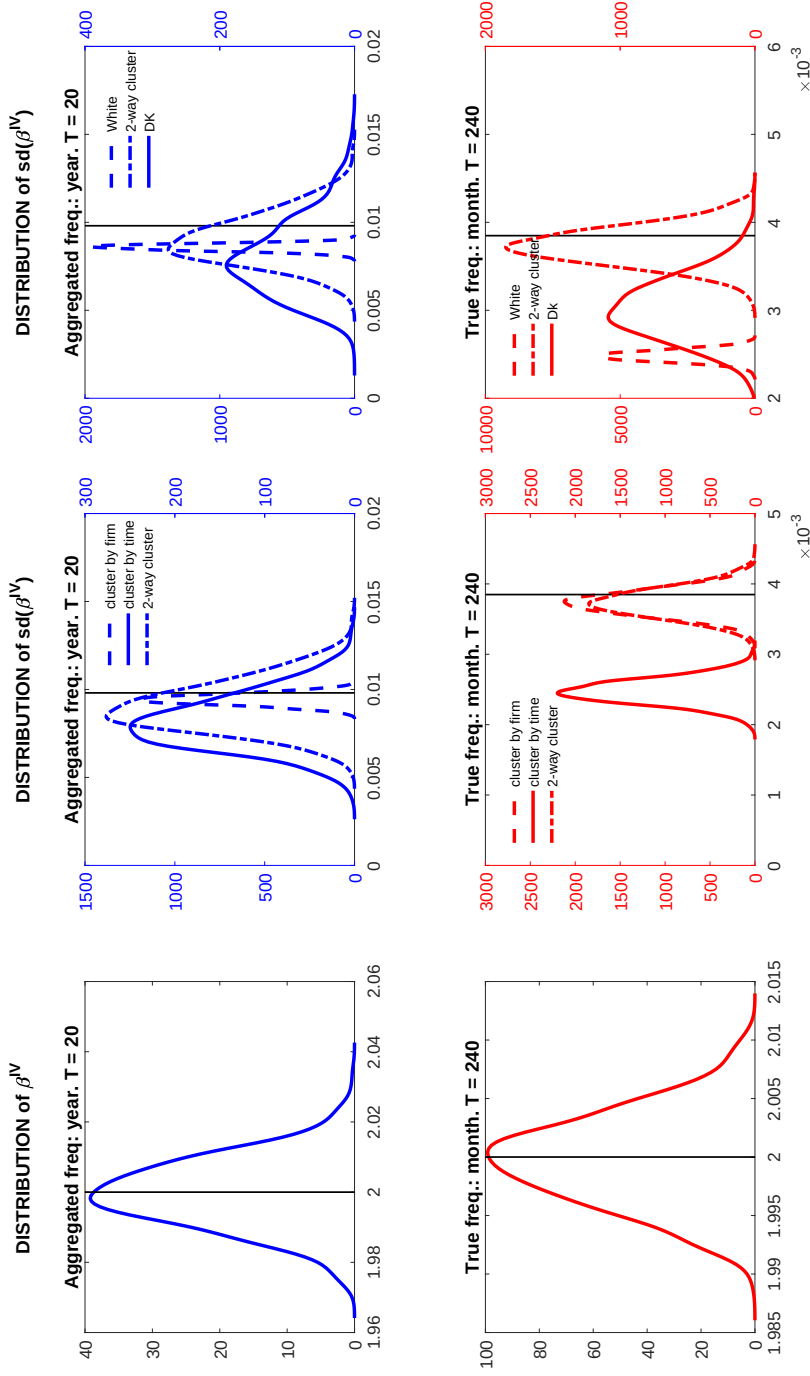
Table 2: PERFORMANCE OF DIFFERENT ESTIMATORS FOR  $SE(\hat{\beta})$   
 ALTERNATIVE SCENARIOS WITH **idiosyncratic regressor**  $x_{i,t}$

| ESTIMATOR                                                                 | ESTIMATED-TO-EMPIRICAL SE |            | 95% CI COVERAGE RATE |            |
|---------------------------------------------------------------------------|---------------------------|------------|----------------------|------------|
|                                                                           | True frequency            | Aggregated | True frequency       | Aggregated |
| PANEL A: SERIAL CORR. ONLY $\max T = 20, N = 2000$                        |                           |            |                      |            |
| H-robust                                                                  | 0.64                      | 0.87       | 0.78                 | 0.92       |
| Firm cluster                                                              | 0.97                      | 0.96       | 0.94                 | 0.95       |
| Time cluster                                                              | 0.64                      | 0.83       | 0.78                 | 0.91       |
| 2-way cluster                                                             | 0.97                      | 0.93       | 0.94                 | 0.93       |
| Driscoll-Kraay                                                            | 0.78                      | 0.82       | 0.87                 | 0.89       |
| PANEL B: SERIAL/SPATIAL CORR. (WEAK) $\max T = 20, N = 2000$              |                           |            |                      |            |
| H-robust                                                                  | 0.12                      | 0.35       | 0.17                 | 0.54       |
| Firm cluster                                                              | 0.19                      | 0.38       | 0.27                 | 0.56       |
| Time cluster                                                              | 0.92                      | 0.76       | 0.94                 | 0.89       |
| 2-way cluster                                                             | 0.93                      | 0.78       | 0.94                 | 0.86       |
| Driscoll-Kraay                                                            | 0.95                      | 0.76       | 0.95                 | 0.87       |
| PANEL C: SERIAL/SPATIAL CORR. (STRONG) $\max T = 20, N = 2000$            |                           |            |                      |            |
| H-robust                                                                  | 0.06                      | 0.12       | 0.10                 | 0.19       |
| Firm cluster                                                              | 0.12                      | 0.13       | 0.20                 | 0.20       |
| Time cluster                                                              | 0.44                      | 0.69       | 0.64                 | 0.83       |
| 2-way cluster                                                             | 0.45                      | 0.69       | 0.64                 | 0.81       |
| Driscoll-Kraay                                                            | 0.72                      | 0.73       | 0.83                 | 0.85       |
| PANEL D: IDIOSYNC. $x_{i,t}$ , SERIAL/SPATIAL CORR. $\max T = 20, N = 50$ |                           |            |                      |            |
| H-robust                                                                  | 0.38                      | 0.79       | 0.55                 | 0.89       |
| Firm cluster                                                              | 0.58                      | 0.85       | 0.76                 | 0.90       |
| Time cluster                                                              | 0.81                      | 0.85       | 0.89                 | 0.91       |
| 2-way cluster                                                             | 0.92                      | 0.91       | 0.93                 | 0.92       |
| Driscoll-Kraay                                                            | 0.88                      | 0.85       | 0.91                 | 0.89       |

Notes: The estimated-to-empirical is computed as the ratio between the mean estimated SE using the estimators proposed and the mean SE computed from the simulated distribution of  $\beta^{IV}$ . The 95% coverage rate presents the percentage of simulation draws for which the estimated 95% confidence interval contained the true  $\beta$ . Each panel presents results for a different set of assumptions regarding the distribution of the error term in equation (18) of the main text.

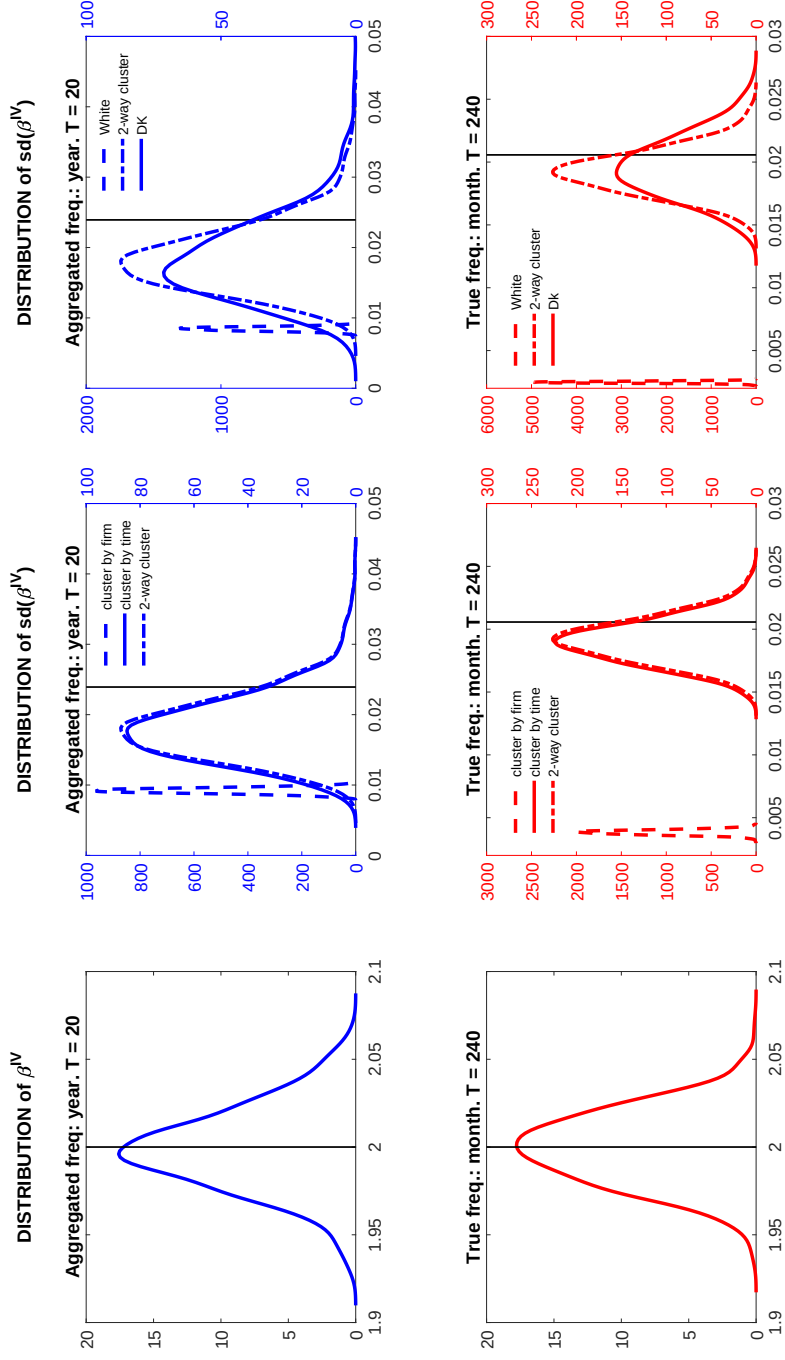


Figure 5: ESTIMATED  $\beta$  AND SE WITH: IDIOSYNCRATIC REGRESSOR; SERIAL CORRELATION IN REGRESSOR AND ERROR.  $N = 2000$ ,  $S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



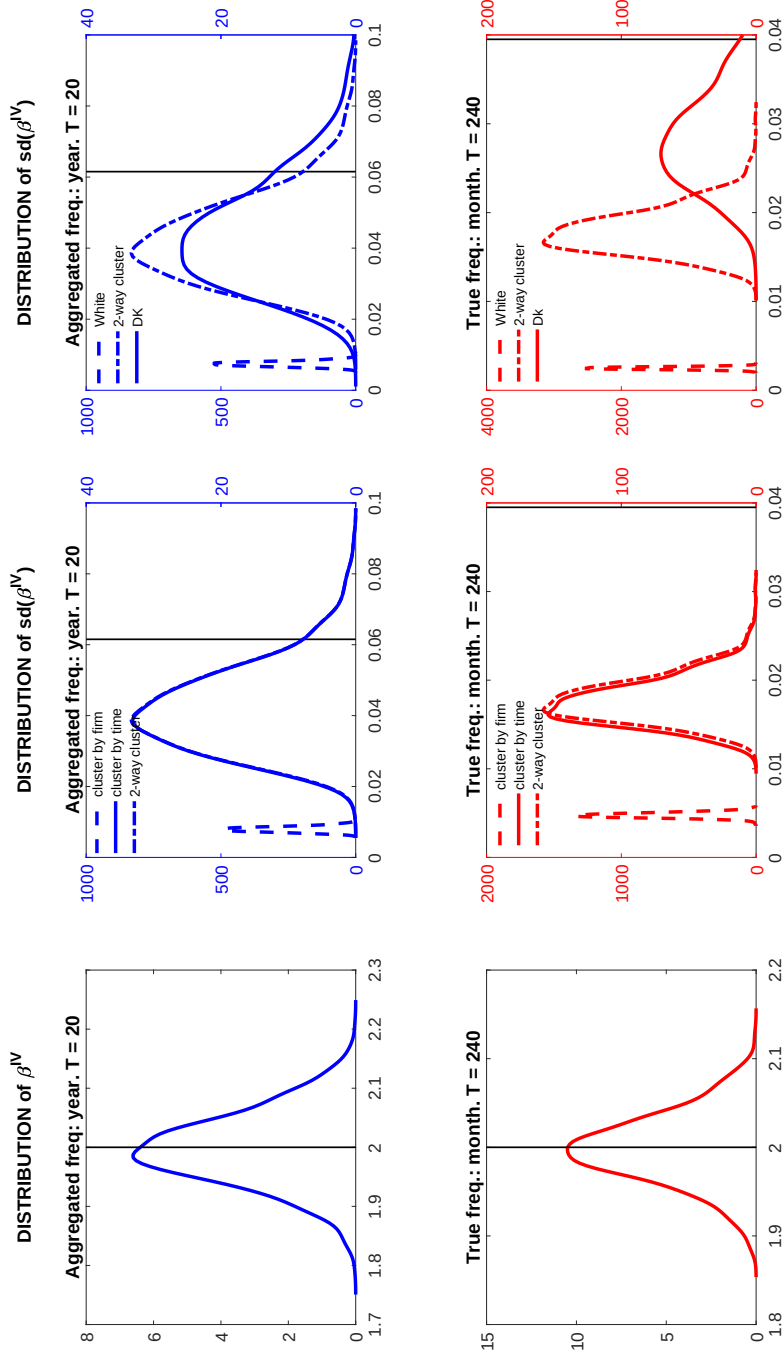
Note: this maps into panel A of table 2. The figures plot the distribution of estimated  $\beta^V$  and  $SE(\beta^V)$  emerging from  $S$  simulated samples from an alternative DGP (22), where the regressor is idiosyncratic. The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation in the regressor and error are:  $\rho_\eta = \rho_\eta^x = 0.85$ ,  $b = 0$ .

Figure 6: ESTIMATED  $\beta$  AND SE WITH: IDIOSYNCRATIC REGRESSOR; SERIAL/ WEAK NON-CONTEMPORANEOUS SPATIAL CORRELATION IN REGRESSOR AND ERROR.  $N = 2000$ ,  $S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



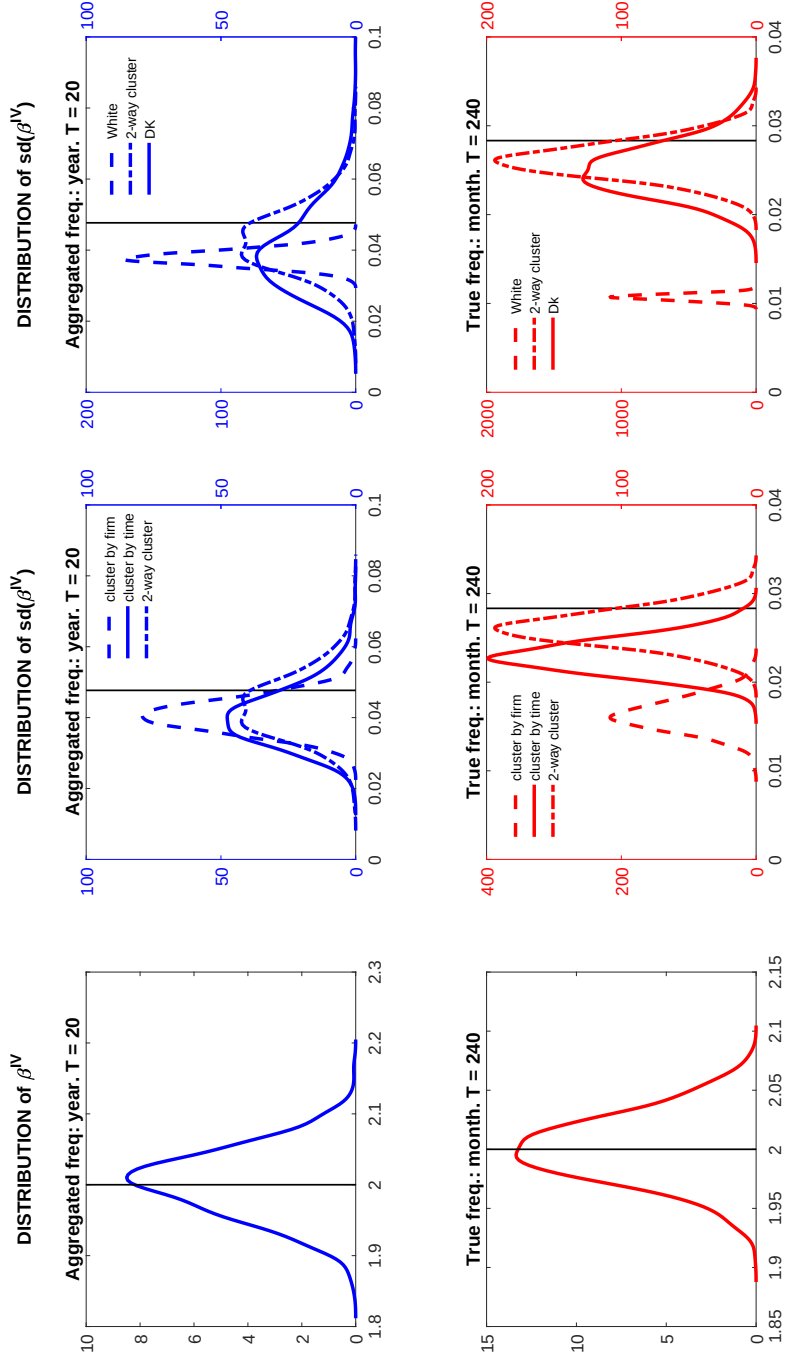
Note: this maps into panel B of table 2. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from an alternative DGP (22), where the regressor is idiosyncratic. The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation in the regressor and error are:  $\rho_\eta = \rho_\eta^x = 0.85$ ,  $\rho_f^x = \rho_f^x = 0.25$ ,  $b = 1$ .

Figure 7: ESTIMATED  $\beta$  AND SE WITH: IDIOSYNCRATIC REGRESSOR; SERIAL/ STRONG NON-CONTEMPORANEOUS SPATIAL CORRELATION IN REGRESSOR AND ERROR.  $N = 2000$ ,  $S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



Note: this maps into panel C of table 2. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from an alternative DGP (22), where the regressor is idiosyncratic. The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation in the regressor and error are:  $\rho_\eta = \rho_\epsilon^x = 0.85$ ,  $\rho_f^x = \rho_f = 0.5$ ,  $b = 1$ .

Figure 8: ESTIMATED  $\beta$  AND SE WITH: IDIOSYNCRATIC REGRESSOR; SERIAL/SPIATAL CORRELATION IN REGRESSOR AND ERROR.  $N = 50$ ,  $S = 1000$  AND MAX  $\tilde{T} = 20$  YEARS



Note: this maps into panel D of table 2. The figures plot the distribution of estimated  $\beta^{IV}$  and  $SE(\beta^{IV})$  emerging from  $S$  simulated samples from an alternative DGP (22), where the regressor is idiosyncratic. The top row shows results from aggregating the monthly frequency from the true DGP into annual data, as described in the main text. The bottom row shows results from estimating the model on the true frequency. The value of the parameters determining the degree of spatial/serial correlation in the regressor and error are:  $\rho_\eta = \rho_\epsilon^x = 0.85$ ,  $\rho_f^x = \rho_f = 0.25$ ,  $b = 1$ .